



AMENI KHABTHANI

AI ENGINEER | VOICE AI SPECIALIST

📍 Mahdia, Tunisia ✉️ amenikhabthani2@gmail.com ☎️ +216 23 513 019
🌐 <https://www.linkedin.com/in/amenikhabthani/> 📄 <https://github.com/Amenikhabthani>

SUMMARY

I am an AI Engineer specializing in NLP, particularly speech technologies and AI development, seeking opportunities to apply my expertise to real-world challenges. I am passionate about developing innovative models; I thrive in collaborative environments and excel at solving complex challenges. I aim to deliver impactful, data-driven solutions that make a meaningful difference.

EXPERIENCE

Data Science Intern

Satoripop, Sousse

6 months - 2024

- Built ASR and TTS solutions, enabling accurate speech processing in French, English, and Tunisian dialects.
- Fine-tuned the Whisper model for Speech Recognition, reducing the word error rate (WER) by 45% on the target dataset
- Fine-tuned XTTS v2 model for Speech Synthesis, improving speech naturalness and intelligibility based on user feedback.
- Built APIs using the FastAPI framework, decreasing response time by 30% and ensuring seamless integration with external applications.
- Designed user-friendly interfaces for APIs with the Gradio toolkit, improving accessibility and usability.

Data Science Intern

Stunas, Sousse

2 months - 2023

- Developed an OCR model for invoice label recognition using DocTr for text detection and NER techniques for label extraction, reducing manual processing time by 50%.

FREELANCE PROJECTS

Creating a Custom TTS System

- Development of a Text-to-Speech (TTS) model based on XTTS v2 fine-tuned from scratch on Maghrebi voice data.
- Assist and support the client in implementing the chain of the solution: data preparation, training, model evaluation.

AI-based Astrology API (End-to-End)

- Design and deployment of an intelligent astrology API combining traditional and modern sources.
- Integration of an LLM with prompt engineering, intent classification, and generation of personalized responses.
- Database management with Supabase for storing user profiles, query histories, and custom configurations.

Seamless Dubbing Automation

- Replaced the existing ElevenLabs-based dubbing service in an API with a custom multilingual dubbing pipeline.
- The new module includes speaker diarization and leverages Whisper Large v3 for ASR, GPT-4 for translation, and hybrid methodology XTTS + Orpheus TTS for TTS synthesis.
- Integrated into a Flask API hosted on Azure App Service
- Handled speech-to-text, speaker segmentation, translation, synthesis, and audio merging to produce aligned, natural-sounding dubbed content.

Medical Text-to-Translation-Text System

- Developed of a text-to-translation-text (T2TT) system for Modern Standard Arabic (MSA) language, designed to preserve domain-specific medical terminology for Arabic-to-English translations, emphasizing clinical and semantic accuracy.
- Dedicated for future integration into healthcare transcription and multilingual medical communication tools.

AI Travel Agent

- Developed an AI-powered voice travel assistant for conversational hotel search and booking using LLM-driven dialogue management and contextual reasoning.
- Implemented an orchestration pipeline integrating intent detection, entity extraction, conversational memory, slot filling, and response generation for natural interactions.
- Integrated Whisper ASR, Groq LLMs, hotel search and weather APIs, and TTS synthesis within a FastAPI backend, alongside responsive web interfaces for real-time voice and text interactions, including authentication and database management.

EDUCATION

MASTER'S DEGREE

Pristini School of AI Oct-2022 / Oct-2024

- Data science, AI, and HealthTech

BACHELOR'S DEGREE

Iset Mahdia Sep-2019 / Jun-2022

- Electrical Engineering Specializing in Automation and Industrial Computing.

CERTIFICATIONS

- | | |
|---|---|
| • Natural Language Processing Specialization (Link) | • Generative AI Engineering with LLMSpecialization (Link) |
| • Generative AI with Large Language Models (Link) | • AWS Academy Cloud Foundations (Link) |
| • Python data structures (Link) | • Data Analysis with Python (Link) |
| • Deeplearning.AI Specialization (Link) | |

SKILLS

- Machine Learning • Deep Learning • Finetuning (PEFT,QLORA,Supervised ft)• Tensorflow • PyTorch • LangChain
- Transformers • LLMs • Generative AI • AI agents • TTS • ASR • VAD • Python • FastAPI • Database management
- Git • Docker • CI/CD deployment • Communication • Problem-Solving • AI Engineering

LANGUAGE

Arabic (native), English (Professional),
French (Conversational)